Rock Paper Scissors Game

Veda Technology Internship | Level 1 | Task 24 | Web Development Track

A rock-paper-scissors game against the computer with a running score. Built with HTML5, CSS3 and vanilla JavaScript.

Live Demo

- GitHub Pages: `https://your-username.github.io/rock-paper-scissors/`
- Repository: `https://github.com/your-username/rock-paper-scissors`

Objective

Practise conditional logic, random choice and simple game state.

Features

- **Three choice buttons**: Rock, Paper, Scissors (also keys `R`, `P`, `S`)
- **Computer's random choice** using `Math.random()` and an array
- **Win / lose / draw logic** in one small function
- **Running score** for You, Draws and Computer, with a short bump animation
- **Reset score** button (also cancels a round in progress)
- Round counter, a result banner (green, red or amber), a highlight on the winning hand, and the last 5 rounds
- A short "Computer is choosing..." pause; buttons are locked during it so clicks cannot double-count
- Accessible: real buttons with labels, results announced with `role="status"`, visible focus, reduced-motion support
- Responsive from 320 px upwards

Tech Stack

Technology	Purpose
HTML5	Buttons, scoreboard, live result area
CSS3	Layout, colour states, animations
JavaScript (ES6)	Random choice, comparison function, game state

Project Structure

```
rock-paper-scissors/
├── index.html
├── style.css
├── script.js
├── README.md
├── Task_24_Report.docx
└── Task_24_Report.pdf
```

How It Works

Random choice

```
function getComputerChoice() {
  return CHOICES[Math.floor(Math.random() * CHOICES.length)];
}
```

`Math.random()` returns a number from 0 up to (not including) 1. Multiplying by 3 and rounding down gives 0, 1 or 2 with equal chance, which is used as an array index.

Deciding the winner

```
const BEATS = { rock: "scissors", paper: "rock", scissors: "paper" };

function getWinner(player, computer) {
  if (player === computer) return "draw";
  return BEATS[player] === computer ? "player" : "computer";
}
```

Only the three winning pairs are written down. Anything else that is not a tie must be a loss, so there are no nine-way `if` chains.

Handling a tie: equal choices return `"draw"`. The Draws counter goes up and neither player's score changes.

Test Cases

Player	Computer	Result
Rock	Scissors	You win
Rock	Paper	You lose
Rock	Rock	Draw
Paper	Rock	You win
Paper	Scissors	You lose
Scissors	Paper	You win
(all 9 combinations tested)		3 wins, 3 losses, 3 draws
300,000 computer picks		rock 33.2%, paper 33.3%, scissors 33.5%
Click during the 0.7 s pause		Ignored
Reset during a round		Score cleared, round cancelled

How to Run

1. Clone: `git clone https://github.com/your-username/rock-paper-scissors.git`
2. Open `index.html` in a browser

Learning Outcomes

- Choosing a random array item
- Writing a clear comparison function with a data table instead of many `if` statements
- Keeping game state in one object and drawing the screen from it
- Locking input during a timed step and cancelling timers on reset

Author

Mantu Yadav, Web Development Intern, Veda Technology